# CHARACTERIZATIONS OF POINTS OF CODIMENSION ZERO ON A QUASI-SEPARATED ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-over-fields.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the identification of the first two conditions by definition, the elementary comparison with generic points through the filtration of a quasi-separated decent space and the standard open neighbourhood on which the chart is an isomorphism, and the final appeal to the proposition on dense open subschemes.

*Editorial note.* The label lemma-generic-point-in-schematic-locus names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Editorial note.* Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source spaces-over-fields.tex –label lemma-generic-point-in-schematic-locus –start 1597 –end 1653 –local-id 28057 returned RESULT=OK (L28057 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-spaces-over-fields/candidates-halfA.json.

*Editorial note.* The proof notes that a more conceptual proof through decent spaces is available and then supplies 'a more elementary proof of the equivalence', which is delivered in full, so the statement is counted.

*Editorial note.* The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

*Allowed standard prerequisites (editorial).*

- Concise description of a quasi-separated decent space. Decent Spaces, the lemma providing a filtration by open subspaces with bijective maps from quasi-compact schemes used in the elementary proof. Supplies the chain of open subspaces and the maps f_i whose points compute the given point.
- Generic points of schemes and constructible subsets. Properties, the lemma that a generic point of a scheme has an open neighbourhood disjoint from a closed subset not containing it, and Same chapter, the lemma describing codimension zero points of a scheme as generic points of irreducible components. Supply the open neighbourhood W on which the map f_i is an isomorphism and the identification of codimension zero points with generic points.
- The open dense subscheme of a locally quasi-separated space. Properties of Spaces, the proposition producing a dense open subspace which is a scheme containing every point of codimension zero. Supplies the final statement of the lemma.

*Date*: stacks-project:spaces-over-fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 00EV: sources/stacks-chow/tags.txt, source lines 549–549. Prerequisite pointer invoked at proof line(s) 1640; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0ABV: sources/stacks-chow/tags.txt, source lines 13033–13033. Prerequisite pointer invoked at proof line(s) 1612; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 07ST: sources/stacks-chow/tags.txt, source lines 9913–9913. Prerequisite pointer invoked at proof line(s) 1629; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original section 03I7: sources/stacks-chow/tags.txt, source lines 4593–4593. Prerequisite pointer invoked at proof line(s) 1614; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0AAW: sources/stacks-chow/tags.txt, source lines 12999–12999. Prerequisite pointer invoked at proof line(s) 1638; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original definition 04NA: sources/stacks-chow/tags.txt, source lines 6071–6071. Prerequisite pointer invoked at proof line(s) 1621; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0BAQ: sources/stacks-chow/tags.txt, source lines 14218–14218. Prerequisite pointer invoked at proof line(s) 1633; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original proposition 06NH: sources/stacks-chow/tags.txt, source lines 8536–8536. Prerequisite pointer invoked at proof line(s) 1652; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *Let $S$ be a scheme. Let $X$ be a quasi-separated algebraic space over $S$. Let $x \in |X|$. The following are equivalent*

1. *$x$ is a point of codimension $0$ on $X$,*
2. *the local ring of $X$ at $x$ has dimension $0$, and*
3. *$x$ is a generic point of an irreducible component of $|X|$.*

*If true, then there exists an open subspace of $X$ containing $x$ which is a scheme.*

*Proof.* The equivalence of (1), (2), and (3) follows from Decent Spaces, Lemma 0ABV and the fact that a quasi-separated algebraic space is decent (Decent Spaces, Section 03I7). However in the next paragraph we will give a more elementary proof of the equivalence.

Note that (1) and (2) are equivalent by definition (Properties of Spaces, Definition 04NA). To prove the equivalence of (1) and (3) we may assume $X$ is quasi-compact. Choose

$$\emptyset = U_{n+1} \subset U_n \subset U_{n-1} \subset \ldots \subset U_1 = X$$

and $f_i : V_i \to U_i$ as in Decent Spaces, Lemma 07ST. Say $x \in U_i$, $x \notin U_{i+1}$. Then $x = f_i(y)$ for a unique $y \in V_i$. If (1) holds, then $y$ is a generic point of an irreducible

component of $V_i$ (Properties of Spaces, Lemma 0BAQ). Since $f_i^{-1}(U_{i+1})$ is a quasi-compact open of $V_i$ not containing $y$, there is an open neighbourhood $W \subset V_i$ of $y$ disjoint from $f_i^{-1}(V_i)$ (see Properties, Lemma 0AAW or more simply Algebra, Lemma 00EV). Then $f_i|_W : W \to X$ is an isomorphism onto its image and hence $x = f_i(y)$ is a generic point of $|X|$. Conversely, assume (3) holds. Then $f_i$ maps $\overline{\{y\}}$ onto the irreducible component $\overline{\{x\}}$ of $|U_i|$. Since $|f_i|$ is bijective over $\overline{\{x\}}$, it follows that $\overline{\{y\}}$ is an irreducible component of $U_i$. Thus $x$ is a point of codimension 0.

The final statement of the lemma is Properties of Spaces, Proposition 06NH. □